

An angular shrinkage BERT model for few-shot relation extraction with none-of-the-above detection[☆]

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ARTICLE INFO

Article history:

Received 8 June 2022

Revised 1 November 2022

Accepted 8 January 2023

Available online 11 January 2023

Edited by: Prof. S. Sarkar

Keywords:

Few-shot learning

Relation extraction

None-of-the-above detection

ABSTRACT

Few-shot relation extraction aims to solve the problem of insufficient annotated data in relation extraction tasks. Through the comparison between samples, few-shot relation extraction achieves lower-cost relation classification. However, most existing methods only do classification within the scope of enumerated relations. For one of the main challenges faced by the application of few-shot relation extraction-the recognition of the none-of-the-above instances, there has been few works on it. In this paper, we propose an angular shrinkage BERT model for the few-shot relation extraction task with none-of-the-above detection, which uses an additive angular loss to enlarge the margins of different classes in the feature space, and obtain highly discriminative features to improve the recognition ability for none-of-the-above instances. Meanwhile, we present a two-stage training strategy to enhance the stability of the performance. We evaluate our model on the most used few-shot relation extraction dataset FewRel. Experimental results show that our approach outperforms previous sentence-pair methods in scenarios containing none-of-the-above instances, and also achieves improvements on the traditional few-shot relation extraction task compared with our baseline model.

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1. Introduction

Relation extraction (RE) is considered as one of the most important topics in the field of information extraction, which is dedicated to identify the relation between two entities in the corpus. Remarkable works have been done for supervised RE task recently [14,15,21,25], which usually obtains samples of each category directly from the training data. However, due to the high cost of obtaining sufficient annotated data in practical application scenarios, few-shot learning is gaining more and more attention in related fields. Inspired by the strong ability of humans to acquire experience from past and apply it to new learning task, few-shot learning was proposed and quickly became a feasible means to accomplish various tasks. Generally speaking, few-shot learning combines a small number of new samples with the experience of the old tasks to achieve great results in the new task.

Few-shot RE models recognize relations by judging whether the relation in the target instance and the relations in the sampled annotated instances are the same. Unlike supervised RE, which learns the representation of each relation from a great quantity of ex-

amples, few-shot RE concentrate on learning a general criterion to classify relations from the training dataset so that the model can be used to deal with relations it has not learned before. Previous works [4,14,22,23] has made great progress in standard few-shot RE task. But they handle the task in an ideal way, assuming that the correct relation must be among the sampled ones. However, in actual application, for an entity pair in the corpus fetched randomly, the actual relation category might be out of the current listed samples or even all the categories in the support set. In order to fix the defect, few-shot none-of-the-above detection (few-shot NOTA) was proposed by Gao et al. [5]. NOTA detection requires detection of an additional NOTA class, which means the class of relation is different from all the listed N classes. To realize NOTA detection, BERT-PAIR [5] treat NOTA as a category like other relations by taking the lowest NOTA confidence in all categories as the final confidence of it. Based on BERT-PAIR, we focus our work on enhancing the model's ability to distinguish NOTA and improving the accuracy of relation classification.

The challenge of NOTA detection is reflected in the difficulty of setting a unified threshold for all relations to determine “not this relation”, especially for relations that the model has not learned before. The methods that compare the vectors processed by network models, represented by Prototypical network [13], gather the vector representations of the same relation together and keep the ones of different relations away from each other. Due to the

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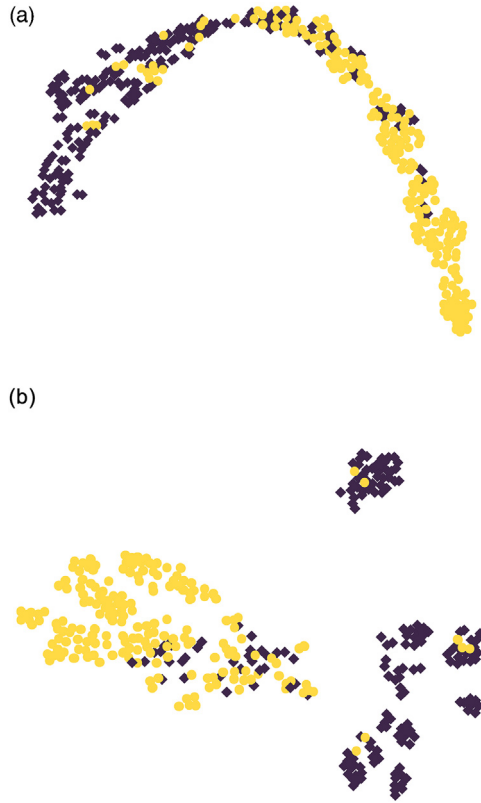


Fig. 1. Comparison on embeddings of 400 sampled sentence pairs between BERT-PAIR and our ASBP model on test datasets. The proposed ASBP model enlarges the margin of NOTA and non-NOTA classes in the embedding space. These embeddings are dimensionally reduced by t-SNE [9]. The NOTA category is marked in yellow and the non-NOTA category is marked in purple.

discrepancy in convergence of each relation's vector representations in this approach, relying solely on vector similarity between instances to identify NOTA tends to be unreliable. Proposed by Gao et al. [5], BERT-PAIR directly inputs sentence pairs into the model and judges from the output whether they have the same relation (non-NOTA) or belong to NOTA, turns NOTA detection into a binary classification task. It leaves the comparison work to the attention network, which makes the pattern of comparison learnable. But as every sample may exist in sentence pairs from both NOTA and non-NOTA categories, the classification boundary of the two categories becomes more ambiguous than conventional tasks, which limits the performance in NOTA detection.

In this paper, we propose a novel Angular Shrinkage BERT-PAIR model (ASBP) for few-shot relation extraction, with a NOTA additive angular (NAA) loss to enhance the discrimination of NOTA, as the performance of few-shot learning methods depends strongly on the distribution of features. Specifically, as shown in Fig. 1, we widen the classification boundary between the NOTA and non-NOTA features by improving the convergence ability of the classifier. Meanwhile, we replace the CLS-token output embedding used in previous work with the average of BERT embeddings which is assumed to be more effective in semantic textual similarity tasks [11]. Then we present a two-stage training strategy to alleviate the self-interference in the training process of few-shot NOTA. Comparative experiments demonstrate the effectiveness and rationality of our method.

The main contributions of our proposed method can be summarized as:

- (1) We propose a NOTA additive angular loss for few-shot NOTA tasks, which obtains classification confidences by angular com-

parison of features and enlarges the classification boundary. The loss function is able to optimize the distribution of features and enhance the discrimination of NOTA.

- (2) We propose a 2-stage training strategy to treat the training of NOTA detection as an incremental learning task for traditional few-shot relation extraction. The training strategy improves the performance of non-NOTA classification in the few-shot NOTA task.
- (3) We present a novel Angular Shrinkage BERT model¹ which achieves a 2% improvement over previous sentence-pair methods on the few-shot NOTA task and obtains competitive results against models using external information on the few-shot RE task without any external training data.

2. Related works

2.1. Supervised relation extraction

Supervised RE models mainly work on the premise of sufficient labeled data. Deep learning has been widely used in this task recently. Many kinds of neural models are proposed such as neural graph-based models [27], capsule network-based methods [26] and the most used BERT [3] based models. The research on BERT-based supervised binary RE mainly focuses on the processing of BERT input and output. Wu and He et al. [21] and Zhao et al. [28] mark the location of entities by special tokens and do classification by the concatenation of CLS-token output and average of entity embeddings. Soares et al. [14] introduce the entity marker tokens and use the start embedding of entities for RE. The method is adopted and updated by subsequent works. Zhong and Chen [29] propose typed markers which incorporate entity type information into the markers. Then, in order to further improve the performance of RE, syntactic information is integrated into the model to learn the text structure knowledge [1,15].

2.2. Few-shot relation extraction

Few-shot RE models can be roughly divided into two categories: no external data and using external data. For the former, the training data is restricted to the specified data set. As a widely employed few-shot learning method, Prototypical network [13] is applied to few-shot RE by making classification according to the Euclidean distance between each query instance and the prototype of each category. Based on Prototypical network, Gao et al. [4] and Ye and Ling [24] add the attention mechanism to optimize the comparison strategy between instances. Wen et al. [20] integrates a transformer module into Prototypical network to enhance relation-level feature extraction. Gao et al. [5] pairs up sentences to give a binary classification result, also known as the best model in the first category so far.

Models in the second category work with external databases or additional training data. Soares et al. [14] relies on a large amount of data to learn a unified relation representation to achieve remarkable performance in few-shot RE tasks. Yang et al. [22] introduces textual descriptions of entities and relations obtained from Wikidata to enhance the prototype network. Yang et al. [23] predicts relations by comparing the concept embeddings obtained from databases with word embeddings in the instances.

2.3. Loss function optimization

Based on Softmax loss, in the field of face recognition in recent years, the optimization methods of loss function has been continu-

¹ Our code is publicly available at <https://github.com/Suejunw/ASBP>.

ously proposed to improve the classification effect. Liu et al. [8] expands the decision margin by multiplying the vector angle of correct classification. Then they maps the feature to the hypersphere by regularizing the classification vector and reducing the bias to zero [7]. Ranjan et al. [10] and Wang et al. [18] normalize the feature vector and add a scale layer after then. Wang et al. [19] and Wang et al. [17] add a penalty value to the cosine value to fix the decision margin. Deng et al. [2] applies the penalty value directly to the angle to achieve a more intuitive control of the angle between classes.

3. Method

In this section, we first define the few-shot RE task that includes NOTA detection in Section 3.1 and then we detail our approach in Section 3.2.

3.1. Problem definition

Few-shot RE learns a classifier with better generalization from data examples and then applies it to new tasks with only a few annotated data. Before the process of relation extraction, instances are selected from the annotated data as the support set. The model classifies instances in the query set by referring to the examples provided in the support set. And NOTA refers to the case that the category to which the instance belongs does not exist in the support set.

Specifically, the input of the task consists of a query instance q_i and a support set S_i . And the purpose of the task is to obtain the relation between the entity pair in q_i according to the comparison between q_i and the instances in S_i . The instance $q_i = (x_i, e_{h_i}, e_{t_i})$ refers to the sample to be identified in the task, represents a sentence x with marked head entity e_h and tail entity e_t . And S_i is composed of samples from N relation categories, each of which contains K instances from the corresponding relation. So S_i is defined as:

$$\begin{aligned} S_i = \{ & r_1 : \{(x_{11}, e_{h11}, e_{t11}), \dots, (x_{1K}, e_{h1K}, e_{t1K})\}, \\ & r_2 : \{(x_{21}, e_{h21}, e_{t21}), \dots, (x_{2K}, e_{h2K}, e_{t2K})\}, \dots, \\ & r_N : \{(x_{N1}, e_{hN1}, e_{tN1}), \dots, (x_{NK}, e_{hNK}, e_{tNK})\}, \end{aligned} \quad (1)$$

where $\{r_1, r_2, \dots, r_N\}$ represent N predefined relations between e_h and e_t . Let R denote all of the relations in the datasets. The task is, for every pair of q_i and S_i : $r_n \in R$, to match an $r_n \in \{r_1, r_2, \dots, r_N\}$ for q_i or identify q_i as NOTA. The output of the task is $O_r = \{(q_i, r_i) : r_i \in \{r_1, r_2, \dots, r_N, \text{NOTA}\}\}$.

To train and test evenly, query instances are integrated into query sets according to a predefined number. A query set Q_i is defined as:

$$\begin{aligned} Q_i = \{ & r_1 : \{(x_{11}^*, e_{h11}^*, e_{t11}^*), \dots, (x_{1Y}^*, e_{h1Y}^*, e_{t1Y}^*)\}, \dots, \\ & r_N : \{(x_{N1}^*, e_{hN1}^*, e_{tN1}^*), \dots, (x_{NY}^*, e_{hNY}^*, e_{tNY}^*)\}, \\ & r_1^{\text{nota}} : \{(x_{11}^{\text{nota}}, e_{h11}^{\text{nota}}, e_{t11}^{\text{nota}}), \dots, (x_{1Y}^{\text{nota}}, e_{h1Y}^{\text{nota}}, e_{t1Y}^{\text{nota}})\}, \dots, \\ & r_M^{\text{nota}} : \{(x_{M1}^{\text{nota}}, e_{hM1}^{\text{nota}}, e_{tM1}^{\text{nota}}), \dots, (x_{MY}^{\text{nota}}, e_{hMY}^{\text{nota}}, e_{tMY}^{\text{nota}})\}, \end{aligned} \quad (2)$$

where $\{r_1, r_2, \dots, r_N\}$ are the same as the relations in the corresponding S_i . And r_m^{nota} indicate NOTA relations which can be represented as $r_m^{\text{nota}} \in R$, $r_m^{\text{nota}} \notin \{r_1, r_2, \dots, r_N\}$. The value of M depends on the current NOTA rate. For example, $M = N$ when NOTA rate is 50%. Fig. 2 shows an episode of the task.

3.2. Our approach

The overview of our model is shown in Fig. 3, which describes the classification process of a single query instance. In this sec-

tion, we analyze our approach in terms of embedding representation and classification methods, respectively.

3.2.1. Sentence pair representation

Given an instance (x_i, e_{h_i}, e_{t_i}) from the query set and an instance $(x_{ik}, e_{h_{ik}}, e_{t_{ik}})$ from the support set. To obtain the vector representation for relation judging, following Gao et al. [5], we concatenate x_{ik} and x_i marked with corresponding entities $\{e_{h_{ik}}, e_{t_{ik}}, e_{h_i}, e_{t_i}\}$. Entity markers [14] are inserted at both sides of the entities. Meanwhile, a special classification token '[CLS]' and a separation token '[SEP]' are employed to collect sentence semantic information and demarcate sentence boundaries. The following example shows the marked sentence pair.

Example 1. "[CLS] [E1s] New York [E1e] has an [E2s] airport [E2e], [SEP] [E1s] Peter [E1s] owns a [E2s] camera [E2e]."

Then it is converted to embeddings according to the pre-trained token vector table [27]. We utilize BERT as the encoder of sentence pair embeddings. The main structure of BERT is a multi-layer bidirectional transformer encoder [16], which is pre-trained on corpus from BooksCorpus and Wikipedia. The output of BERT integrates the lexical and semantic information of the sentence. Different from BERT-PAIR, we employ the average of BERT output embeddings as the sentence pair representation, which is considered to have a stronger association with the gold labels for textual similarity tasks [3]. For every query instance q_i , pair each instance in the corresponding support set with it to obtain a sentence pair representation set T_i . Let t_{nk} indicate the sentence pair representation and $[h_0, h_1, \dots, h_{len}]$ denote the output of BERT, T_i is represented as:

$$T_i = \{ \{t_{11}, t_{12}, \dots, t_{1K}\}, \{t_{21}, t_{22}, \dots, t_{2K}\}, \dots, \{t_{N1}, t_{N2}, \dots, t_{NK}\} \}, \quad (3)$$

$$t_{nk} = \text{Avg}([h_0, h_1, \dots, h_{len}]_{nk}). \quad (4)$$

3.2.2. Relation classification

After the sentence pair representation is obtained, a relation classifier is required to determine which category the query sentence belongs to. Previous works [4,5,14,22,23] use the softmax-classifier for relation classification. As the most widely used loss function for classification, Softmax loss is represented as:

$$L_1 = -\frac{1}{B} \sum_{i=1}^B \log \frac{e^{W_j^T t_i + b_{y_i}}}{\sum_{j=1}^N e^{W_j^T t_i + b_j}}, \quad (5)$$

where $t_i \in \mathbb{R}^d$ indicate the vector representation of the i th sample, which belongs to the y_i th class. B denotes the batch size. $W_j \in \mathbb{R}^d$ refers to the j th column of the weight matrix $W \in \mathbb{R}^{d \times n}$ and b_j is the bias. However, the difference between training data and predicted data is relatively more obvious in few-shot RE tasks. And the softmax-classifier cannot handle this distinction effectively. More specifically, some instances may fall into the wrong classification region due to the distinction and the ambiguity of classification boundaries. The situation calls for classification strategies with higher convergence. Insightface loss [2] normalizes the feature t_i and the weight matrix W to measure the similarity of features only by angle difference and map the vectors onto the hypersphere. Then it adds an additive value m to the angle, which can also be considered as a penalty value, to force the features to converge. It is represented as follows:

$$L_2 = -\frac{1}{B} \sum_{i=1}^B \log \frac{e^{s(\cos(\theta_{y_i} + m))}}{e^{s(\cos(\theta_{y_i} + m))} + \sum_{j=1, j \neq y_i}^N e^{s \cos \theta_j}}, \quad (6)$$

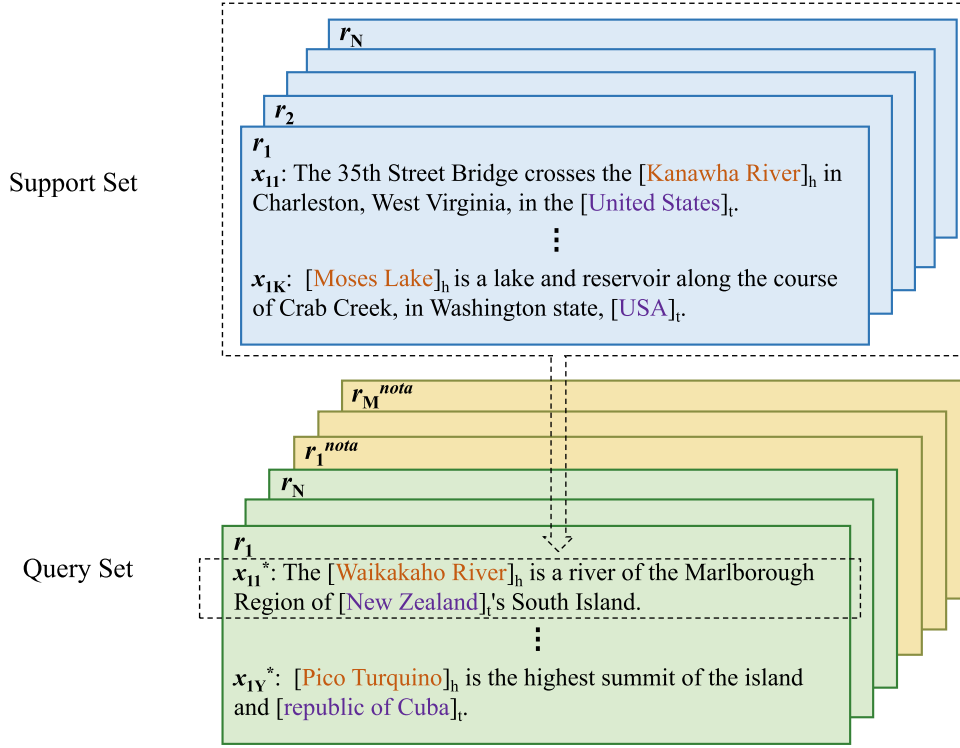


Fig. 2. An episode of few-shot NOTA task, which is composed of a query set and a corresponding support set. The task aims to identify the category to which each instance in the query set belongs, with reference to the support set.

where θ_j denotes the angle between the weight W_j and the feature t_i described in Eq. (5). s is an amplification factor. Insightface loss has effectively improved the classification accuracy in face recognition tasks. We combine its core strategies with few-shot NOTA task and propose a NOTA Additive Angular Loss, which can be represented as:

$$L_3 = -\frac{1}{B} \sum_{i=1}^B \log \frac{e^{c_{y_i}}}{\sum_{j=1}^N e^{c_{ij}} + e^{c_i^{NOTA}}}, \quad (7)$$

where c_{ij} and c_i^{NOTA} express the confidence that the i th instance belongs the j th relation and NOTA, respectively. $c_{y_i} \in \{c_{i1}, c_{i2}, \dots, c_{iN}, c_i^{NOTA}\}$ is the confidence of the class which the i th instance belongs to. c_{ij} and c_i^{NOTA} are calculated as follows:

$$c_{ij} = \frac{1}{K} \sum_{k=1}^K s(\cos(\theta_{ijk} + m)), \quad (8)$$

$$c_i^{NOTA} = \min \{c_{i1}^{NOTA}, c_{i2}^{NOTA}, \dots, c_{iN}^{NOTA}\}, \quad (9)$$

$$c_{ij}^{NOTA} = \frac{1}{K} \sum_{k=1}^K s(\cos(\theta_{ijk}^{NOTA} + m)), \quad (10)$$

where θ_{ijk} and θ_{ijk}^{NOTA} refer to the angle between the feature representation $t_{jk} \in \mathbb{R}^d$ and the weight vectors $W_1 \in \mathbb{R}^d$ and $W_2 \in \mathbb{R}^d$, respectively. W_1 and W_2 form the weight matrix $W \in \mathbb{R}^{d \times 2}$. c_{ij} and c_{ij}^{NOTA} are obtained by averaging the corresponding K confidences. As a separate category, the NOTA confidence c_i^{NOTA} is assigned the minimum of c_{ij}^{NOTA} . Because to predict that none of the N relations match the instance, the key to consider is the relation that best matches it.

3.2.3. Incremental training strategy

Previous works [4,14,22,23] train few-shot RE models on random sampled batched instances. Gao et al. [5] continues this approach and deal NOTA detection task with extra NOTA instances. The training of NOTA detection and the training of few-shot RE without NOTA instances are carried out simultaneously. The former is a binary classification task in BERT-PAIR [5] but the latter is a confidence comparison task under the same class. We conjecture that these two different types of tasks interfere with each other when performed simultaneously. So, we divide the training of the model into two stages. In the first stage we train the model without NOTA instances so that the model only focuses on learning confidence comparisons for the non-NOTA category. Then we add the NOTA instances to the support set in the second stage to achieve complete few-shot NOTA training. In this way, we ensure the model's ability to classify non-NOTA instances, thus making the experimental results more stable. The experimental results show that this two-stage training method bring a certain improvement to the results of each sub-experiment.

4. Experiment

4.1. Setup

Dataset We evaluate our approach on the most commonly used few-shot RE dataset FewRel [5,6]. The FewRel dataset extract 70,000 instances from Wikipedia that can be divided into 100 relations equally. Each instance contains a sentence and the location of two entities. Among the dataset, the test set, which has 20 relations, is not made public. So we follow previous works [22,23] to re-split the published 80 relations into 50, 14 and 16 for training, validation and testing, respectively.

Evaluation The evaluation of few-shot RE is commonly in the form of N -way- K -shot, where N and K denote the number of classes and instances from each class in the support set, respec-

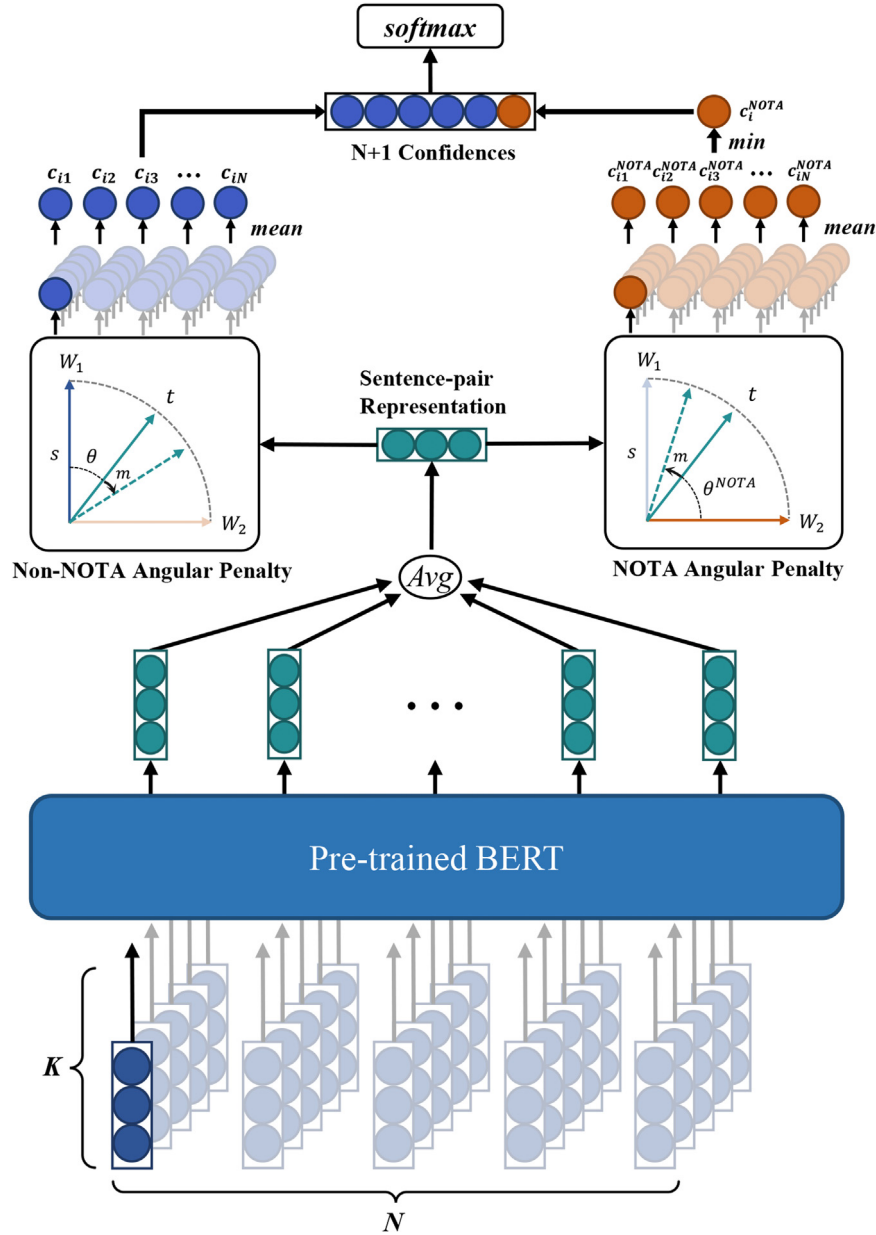


Fig. 3. The framework of ASBP model. The input to the model is $N \times K$ sentence pair embeddings, each of which has its confidences calculated independently. Then, the $N+1$ classification scores of the query instance are obtained through the calculation of the $2 \times N \times K$ confidences.

tively. Our approach is evaluated in different scenarios by setting different values of N and K . Furthermore, following Gao et al. [5], in scenarios that include NOTA detection, our models are tested under different NOTA rates while all the models are trained given 50% NOTA queries. As a standard evaluation protocol for few-shot RE tasks, accuracy is used as the performance metric.

Implementation details We employ bert-base-uncased [3] as the base encoder of our model and the hidden size is 768. Hyperparameters such as learning rate and dropout rate follow the settings in BERT-PAIR [5]. In order to obtain the best value of the additive angle m , which has an impact on the performance, we experimented with 5 values ranging from 0.1 to 0.5. As the results shown in Fig. 4, we set the default value of m to be 0.3. Then, according to Deng et al. [2], the magnification factor s is designated as 64. We measure the performance on the validation set every 500 steps during training to get the best model weights. And we also use the early stopping strategy to prevent overtraining. For

tasks including NOTA detection, all the models are trained under 50% NOTA rate.

4.2. Performance and comparisons

Table 1 compares the performance of our model with the baseline model on few-shot NOTA task. For a fair comparison, we test BERT-PAIR and our model on the re-split dataset. And for all the

Table 1
Comparison of accuracy rates (%) on few-shot NOTA.

		BERT-PAIR	ASBP	ASBP(2-Stages)
5-Way 1-Shot	15% NOTA	72.9	73.75	73.9
	50% NOTA	71.81	73.13	73.87
5-Way 5-Shot	15% NOTA	75.81	77.15	77.86
	50% NOTA	74.69	76.17	76.82

Table 2
Comparison of accuracy rates (%) on few-shot RE.

Model	Encoder	EXT-INFO	5-w-1-s	5-w-5-s	10-w-1-s	10-w-5-s
Proto [13]	CNN		74.29	85.18	61.15	74.41
HATT-Proto [4]	CNN		74.84	85.81	62.05	75.25
MLMAN [24]	CNN		78.21	88.01	65.70	78.35
BERT-PAIR [5]	BERT		82.57	88.47	73.37	81.10
TD-Proto [22]	BERT	✓	84.76	92.38	74.32	85.92
ConceptFERE [23]	BERT	✓	89.21	-	75.72	-
ConceptFERE (Simple)	BERT	✓	84.28	90.34	74.00	81.82
ASBP(ours)	BERT		86.28	91.14	76.84	84.58

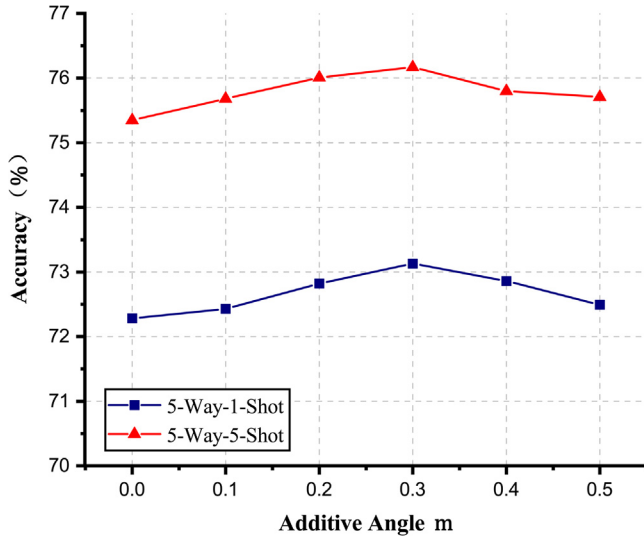


Fig. 4. Test accuracy line graph of ASBP with additive angle changed. Train and test under 50% NOTA rate.

experiments presented in Table 1, we report the average accuracy of 5 runs. The results show that our ASBP model achieves an overall improvement compared with the baseline. More specifically, the directly trained ASBP achieves 0.85%, 1.32%, 1.34%, and 1.48% gains over BERT-PAIR in the given 4 scenarios, respectively. And the ASBP employing the 2-stage training strategy improves these values to 1.00%, 2.06%, 2.05% and 2.13%. This indicates that the first stage of training ensures the performance of the model on non-NOTA classification and makes the final results more stable.

Table 2 tabulates the performance of comparable models on few-shot RE task, where the scores of previous methods [4,5,13,22–24] are derived from Yang et al. [23]. The algorithms are divided into three parts. The methods in the first part do not use external information while the ones in the second part do. The third part lists our method, which is of the same type as the methods in the first part. For fairness of comparison, we choose the best result in 3 runs as our final scores on few-shot RE task. It can be observed that the performance of the methods in the second part is significantly improved compared to the ones in the first part, and our ASBP model achieves competitive performances comparing with them without using any external information. As the few-shot RE task does not involve the distinction of NOTA, such improvements demonstrate the effectiveness of features averaging and hypersphere mapping.

4.3. Ablation study

In this section, to demonstrate the effectiveness of the averaged embedding representation (Avg-E) and NAA loss represented

Table 3
Results of ablation study.

Model	5-w-1-s(50%NOTA)
ASBP(2-Stage)	73.87
ASBP	73.13
w/o AA	72.34
w/o AA & HM	71.94
w/o Avg-E	72.42

in Section 3.2, we state the experimental results with different settings. The NAA loss can be divided into two parts, hypersphere mapping module (HM) and additive angle module (AA). As shown in Table 3, the accuracy of ASBP model drops by 0.7% to 0.8% when removing the additive angle module or employing the first embedding of BERT as the feature representation. Meanwhile, the model without both hypersphere mapping module and additive angle module performs worse. The results proves that the proposed methods can effectively improve the classification convergence and the accuracy of similarity comparison for few-shot RE tasks.

4.4. Analysis and discussion

We have shown the improvement of our model over the previous outstanding models. In this section, we focus on discussing what contributes to the final performance and further exploiting the potential of our model.

As described in Section 3.2, we propose the NAA loss to enhance the ability of classification model to recognize NOTA. Previous works used a fully-connected layer to get the confidence, which multiplies the feature vector by a vector and adds a bias to it. As the product of vectors is related to both the direction and the length of the vectors, the confidence is influenced by three sets of variables at the same time. Through the hypersphere mapping strategy, we exclude the effects of vector length and the bias, so that the angle between the vectors becomes the only factor which determine the confidence. Based on the improvement shown in Table 2, we conjecture that the reduction of these distractors weakens the impact of data deviation in the few-shot RE task.

On the other hand, by adding additive angle, we obtain an improvement in feature separability as shown in Fig. 1. While Deng et al. [2] only adds additive angle to the confidence calculation of the correct category, we do it for all categories. For incorrect categories, the softmax function pushes the confidences to be lower during training and the exponential function has smaller derivatives for smaller values. These limit the effect of additive angle on them and enables our proposed NAA loss to improve the convergence of features. Thus, comparing with Insightface loss, our method simplifies the algorithm without affecting the performance, which can be confirmed by Fig. 1(b). Meanwhile, as shown in Fig. 4, the performance of ASBP model shows a decreasing trend in our experiments when the additional angle is more than 0.3.

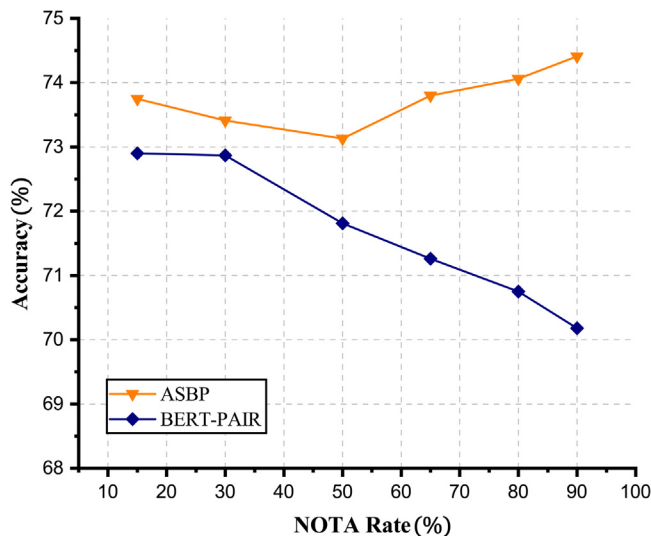


Fig. 5. Accuracy comparison between ASBP and BERT-PAIR with different NOTA rates. All models are trained under 50% NOTA rate. Take the average of 5 runs.

Because the relation classification without NOTA instances is done by comparing confidences of different categories in our method. We believe that an excessively large additional angle will bring too much force for features to converge, thereby disturbing the distribution of feature vectors and impairing the performance.

Combining the above characteristics, our ASBP model achieves significant improvement in distinguishing NOTA. The NOTA rate in the experiment is artificially controlled below 50%. However, according to the statistics presented by Sabo et al. [12], the NOTA rate of data in real scenes exceeds 90%, which brings greater challenges to the few-shot RE tasks. According to the accuracy comparison shown in Fig. 5, our model outperforms BERT-PAIR by 4% at 90% NOTA rate, and maintains stability under different NOTA rates. Then, in order to test the performance of ASBP in realistic task scenarios, we conducted experiments on an unpublished industrial dataset and achieved an average improvement of 1.93% compared to BERT-PAIR. This progress is benefited from the improved NOTA separability described above. Therefore, our proposed method has good potential for real-world tasks, and provides a reference for subsequent research works on few-shot RE tasks.

5. Conclusion

In this paper, we propose an ASBP model for the few-shot relation extraction task including NOTA detection, which optimizes the distribution of feature vectors with a more convergent classification strategy. And we present a two-stage training strategy for few-shot NOTA. Experimental results demonstrate the effectiveness of our method compared to state-of-the-art algorithms in both scenarios with and without NOTA instances. Then we analyze the reasons for the improvement of our method and its advantages and limitations based on the experiments for comparison. In future work, we may concentrate on leveraging self-supervised learning to achieve more fine-grained feature distribution optimization in the few-shot relation extraction task.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This work was supported in part by Science and Technology Innovation 2030 - Major Project of “New Generation Artificial Intelligence” granted by Ministry of Science and Technology (No. 2020AAA0109300), in part by “Science and Technology Innovation Action Plan” of Shanghai Science and Technology Commission for social development project under Grant 21DZ1204900, in part by Shanghai Local Capacity Enhancement project (No. 21010501500).

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